



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Canada WHMIS 2015 which includes the amended Hazardous Products Act (HPA) and the Hazardous Products Regulation (HPR)

Revision date 30-May-2024

Revision Number 1.2

## 1. Identification

### Product identifier

Product Name PureZOL RNA Isolation Reagent

### Other means of identification

Catalog Number(s) 7326890, 7326890EDU, 7326880, 10001480

### Recommended use of the chemical and restrictions on use

Recommended use Laboratory chemicals

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
1329 Meyerside Drive  
Mississauga, ON L5T 1C9  
Canada

#### Technical Service

1-800-361-1808  
support@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

|                                                    |                           |
|----------------------------------------------------|---------------------------|
| Acute toxicity - Oral                              | Category 3                |
| Acute toxicity - Dermal                            | Category 4                |
| Acute toxicity - Inhalation (Dusts/Mists)          | Category 3                |
| Skin corrosion/irritation                          | Category 1 Sub-category B |
| Serious eye damage/eye irritation                  | Category 1                |
| Germ cell mutagenicity                             | Category 2                |
| Specific target organ toxicity (repeated exposure) | Category 2                |
| Corrosive to metals                                | Category 1                |
| Flammable liquids                                  | Category 4                |

### Label elements

#### **Danger**

#### **Hazard statements**

Toxic if swallowed

Harmful in contact with skin  
Toxic if inhaled  
Causes severe skin burns and eye damage  
Suspected of causing genetic defects  
May cause damage to organs through prolonged or repeated exposure  
May be corrosive to metals  
Combustible liquid

**Precautionary Statements - Prevention**

Wear protective gloves/protective clothing/eye protection/face protection  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Do not breathe dust/fume/gas/mist/vapors/spray  
Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Keep only in original packaging

**Precautionary Statements - Response**

IF exposed or concerned: Get medical advice/attention  
Immediately call a POISON CENTER or doctor

**Eyes**

Immediately call a POISON CENTER or doctor  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

**Skin**

IF ON SKIN: Wash with plenty of water and soap  
Call a POISON CENTER or doctor if you feel unwell  
Take off contaminated clothing and wash it before reuse  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower]  
Wash contaminated clothing before reuse

**Inhalation**

IF INHALED: Remove person to fresh air and keep comfortable for breathing  
Call a POISON CENTER or doctor  
Immediately call a POISON CENTER or doctor

**Ingestion**

IF SWALLOWED: Immediately call a POISON CENTER or doctor  
Rinse mouth  
IF SWALLOWED: Rinse mouth. Do NOT induce vomiting

**Fire**

In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Spill**

Absorb spillage to prevent material damage

**Precautionary Statements - Storage**

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Store in a well-ventilated place. Keep container tightly closed  
Store in corrosion resistant container with a resistant inner liner

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

No information available.

**Unknown acute toxicity**

0 % of the mixture consists of ingredient(s) of unknown acute oral toxicity

0 % of the mixture consists of ingredient(s) of unknown acute dermal toxicity

0 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (dust/mist)

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

| Chemical name                                  | CAS No.  | Weight-% | Hazardous Material<br>Information Review Act<br>registry number<br>(HMIRA registry #) | Date HMIRA filed and<br>date exemption granted<br>(if applicable) |
|------------------------------------------------|----------|----------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Phenol                                         | 108-95-2 | 35 - 50  | -                                                                                     |                                                                   |
| Thiocyanic acid, compound with guanidine (1:1) | 593-84-0 | 20 - 35  | -                                                                                     |                                                                   |

**4. First-aid measures****Description of first aid measures****General advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Inhalation**

Remove to fresh air. If breathing has stopped, give artificial respiration. Get medical attention immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. If breathing is difficult, (trained personnel should) give oxygen. Delayed pulmonary edema may occur. Immediate medical attention is required.

**Eye contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Keep eye wide open while rinsing. Do not rub affected area. Get immediate medical advice/attention.

**Skin contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get immediate medical advice/attention.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Never give anything by mouth to an unconscious person. Do NOT induce vomiting. Get immediate medical advice/attention.

**Self-protection of the first aider** Avoid contact with skin, eyes or clothing. Remove all sources of ignition. Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Do not breathe vapor or mist. Use personal protective equipment as required. See section 8 for more information.

#### **Most important symptoms and effects, both acute and delayed**

**Symptoms** Burning sensation. Coughing and/ or wheezing. Difficulty in breathing.

#### **Indication of any immediate medical attention and special treatment needed**

**Note to physicians** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Do not give chemical antidotes. Asphyxia from glottal edema may occur. Marked decrease in blood pressure may occur with moist rales, frothy sputum, and high pulse pressure.

### **5. Fire-fighting measures**

**Suitable Extinguishing Media** Dry chemical. Carbon dioxide (CO<sub>2</sub>). Water spray. Alcohol resistant foam.

**Unsuitable extinguishing media** No information available.

**Specific hazards arising from the chemical** Keep product and empty container away from heat and sources of ignition. In the event of fire, cool tanks with water spray. The product causes burns of eyes, skin and mucous membranes. Thermal decomposition can lead to release of irritating gases and vapors.

#### **Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** Yes.

**Special protective equipment for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

### **6. Accidental release measures**

#### **Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. See section 8 for more information. Take precautionary measures against static discharges. Do not touch or walk through spilled material. Attention! Corrosive material. Keep people away from and upwind of spill/leak. Do not breathe vapor or mist.

**Other information** Refer to protective measures listed in Sections 7 and 8.

#### **Methods and material for containment and cleaning up**

**Methods for containment** Stop leak if you can do it without risk. Do not touch or walk through spilled material. Dike far ahead of liquid spill for later disposal.

**Methods for cleaning up** Take precautionary measures against static discharges. Dam up. Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.

## 7. Handling and storage

### Precautions for safe handling

#### Advice on safe handling

Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Use personal protection equipment. Do not breathe vapor or mist. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Take precautionary measures against static discharges. In case of insufficient ventilation, wear suitable respiratory equipment. Handle product only in closed system or provide appropriate exhaust ventilation. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse. Remove contaminated clothing and shoes.

### Conditions for safe storage, including any incompatibilities

#### Storage Conditions

Keep containers tightly closed in a dry, cool and well-ventilated place. Protect from moisture. Store locked up. Keep out of the reach of children. Store away from other materials. Keep away from heat, sparks, flame and other sources of ignition (i.e., pilot lights, electric motors and static electricity). Keep in properly labeled containers. Store in accordance with the particular national regulations. Store in accordance with local regulations. Store according to product and label instructions.

## 8. Exposure controls/personal protection

### Control parameters

#### Exposure Limits

| Chemical name      | Alberta                                         | British Columbia   | Ontario            | Quebec                                          |
|--------------------|-------------------------------------------------|--------------------|--------------------|-------------------------------------------------|
| Phenol<br>108-95-2 | TWA: 5 ppm<br>TWA: 19 mg/m <sup>3</sup><br>Skin | TWA: 5 ppm<br>Skin | TWA: 5 ppm<br>Skin | TWA: 5 ppm<br>TWA: 19 mg/m <sup>3</sup><br>Skin |

### Appropriate engineering controls

#### Engineering controls

Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

#### Eye/face protection

Tight sealing safety goggles. Face protection shield.

#### Hand protection

Wear suitable gloves. Impervious gloves.

#### Skin and body protection

Wear suitable protective clothing. Long sleeved clothing. Chemical resistant apron.

#### Respiratory protection

No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

#### General hygiene considerations

Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product. Regular cleaning of equipment, work area and clothing is recommended. Contaminated work clothing should not be allowed out of the workplace. Wash hands before breaks and immediately after handling the product. Avoid contact with skin, eyes or clothing. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Do not breathe vapor or mist.

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                |                          |
|----------------|--------------------------|
| Physical state | Liquid                   |
| Appearance     | aqueous solution         |
| Color          | dark red                 |
| Odor           | sweet                    |
| Odor threshold | No information available |

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      | 4-5               |                         |
| Melting point / freezing point          | No data available | None known              |
| Initial boiling point and boiling range | > 100 °C / 212 °F |                         |
| Flash point                             | > 80 °C / 176 °F  |                         |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapor pressure                          | No data available | None known              |
| Relative vapor density                  | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Miscible in water |                         |
| Solubility in other solvents            | No data available | None known              |
| Partition coefficient                   | No data available | None known              |
| Autoignition temperature                | No data available | None known              |
| Decomposition temperature               |                   | None known              |
| Kinematic viscosity                     | No data available | None known              |
| Dynamic viscosity                       | No data available | None known              |
| <u>Other information</u>                |                   |                         |
| Explosive properties                    | Not applicable.   |                         |
| Oxidizing properties                    | Not applicable.   |                         |
| Softening point                         | Not applicable    |                         |
| Molecular weight                        | Not applicable    |                         |
| VOC content                             | Not applicable    |                         |

## 10. Stability and reactivity

|                                    |                                                                                              |
|------------------------------------|----------------------------------------------------------------------------------------------|
| Reactivity                         | No information available.                                                                    |
| Chemical stability                 | Stable under normal conditions.                                                              |
| Possibility of hazardous reactions | None under normal processing.                                                                |
| Conditions to avoid                | Exposure to air or moisture over prolonged periods. Heat, flames and sparks. Excessive heat. |
| Incompatible materials             | Oxidizing agent. Acids. Bases.                                                               |
| Hazardous decomposition products   | None known based on information supplied.                                                    |

## 11. Toxicological information

**Information on likely routes of exposure****Product Information**

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available Corrosive by inhalation (based on components). Inhalation of corrosive fumes/gases may cause coughing, choking, headache, dizziness, and weakness for several hours. Pulmonary edema may occur with tightness in the chest, shortness of breath, bluish skin, decreased blood pressure, and increased heart rate. Inhaled corrosive substances can lead to a toxic edema of the lungs. Pulmonary edema can be fatal. Toxic by inhalation.              |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available Causes serious eye damage (based on components). Corrosive to the eyes and may cause severe damage including blindness. May cause irreversible damage to eyes.                                                                                                                                                                                                                                                                                         |
| <b>Skin contact</b> | Specific test data for the substance or mixture is not available Corrosive (based on components). Causes burns. May be absorbed through the skin in harmful amounts. Harmful in contact with skin.                                                                                                                                                                                                                                                                                                                      |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available Causes burns (based on components). Ingestion causes burns of the upper digestive and respiratory tracts. May cause severe burning pain in the mouth and stomach with vomiting and diarrhea of dark blood. Blood pressure may decrease. Brownish or yellowish stains may be seen around the mouth. Swelling of the throat may cause shortness of breath and choking. May cause lung damage if swallowed. May be fatal if swallowed and enters airways. |

**Symptoms related to the physical, chemical and toxicological characteristics**

**Symptoms** Redness. Burning. May cause blindness. Coughing and/ or wheezing. Difficulty in breathing.

**Acute toxicity****Numerical measures of toxicity**

No information available

**The following values are calculated based on chapter 3.1 of the GHS document**

|                                      |                |
|--------------------------------------|----------------|
| <b>ATEmix (oral)</b>                 | 171.00 mg/kg   |
| <b>ATEmix (dermal)</b>               | 1,260.00 mg/kg |
| <b>ATEmix (inhalation-dust/mist)</b> | 0.835 mg/l     |

**Unknown acute toxicity**

- 0 % of the mixture consists of ingredient(s) of unknown acute oral toxicity
- 0 % of the mixture consists of ingredient(s) of unknown acute dermal toxicity
- 0 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (dust/mist)

**Component Information**

| Chemical name      | Oral LD50           | Dermal LD50            | Inhalation LC50 |
|--------------------|---------------------|------------------------|-----------------|
| Phenol<br>108-95-2 | = 340 mg/kg ( Rat ) | = 630 mg/kg ( Rabbit ) | -               |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                                                                                                       |
|------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Classification based on data available for ingredients. Causes burns.                                 |
| <b>Serious eye damage/eye irritation</b> | Classification based on data available for ingredients. Risk of serious damage to eyes. Causes burns. |
| <b>Respiratory or skin sensitization</b> | Based on available data, the classification criteria are not met.                                     |

**Germ cell mutagenicity** Contains a known or suspected mutagen. Classification based on data available for ingredients. Suspected of causing genetic defects.

**Carcinogenicity** Based on available data, the classification criteria are not met.

The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Chemical name      | ACGIH | IARC    | NTP | OSHA |
|--------------------|-------|---------|-----|------|
| Phenol<br>108-95-2 | -     | Group 3 | -   | -    |

#### Legend

**IARC (International Agency for Research on Cancer)**

Group 3 - Not Classifiable as to Carcinogenicity in Humans

**Reproductive toxicity** Based on available data, the classification criteria are not met.

**STOT - single exposure** Based on available data, the classification criteria are not met.

**STOT - repeated exposure** May cause damage to organs through prolonged or repeated exposure.

**Target organ effects** Liver, Kidney, Respiratory system, Eyes, Skin.

**Aspiration hazard** Based on available data, the classification criteria are not met.

## 12. Ecological information

**Ecotoxicity** Toxic to aquatic life. Toxic to aquatic life with long lasting effects.

| Chemical name      | Algae/aquatic plants                                                                                                                                                              | Fish                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Toxicity to microorganisms | Crustacea                                                                                |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------|
| Phenol<br>108-95-2 | EC50: =46.42mg/L (96h, Pseudokirchneriella subcapitata)<br>EC50: 0.0188 - 0.1044mg/L (96h, Pseudokirchneriella subcapitata)<br>EC50: 187 - 279mg/L (72h, Desmodesmus subspicatus) | LC50: 11.9 - 50.5mg/L (96h, Pimephales promelas)<br>LC50: 20.5 - 25.6mg/L (96h, Pimephales promelas)<br>LC50: =32mg/L (96h, Pimephales promelas)<br>LC50: 5.449 - 6.789mg/L (96h, Oncorhynchus mykiss)<br>LC50: 7.5 - 14mg/L (96h, Oncorhynchus mykiss)<br>LC50: 4.23 - 7.49mg/L (96h, Oncorhynchus mykiss)<br>LC50: 5.0 - 12.0mg/L (96h, Oncorhynchus mykiss)<br>LC50: =13.5mg/L (96h, Lepomis macrochirus)<br>LC50: 11.9 - 25.3mg/L (96h, Lepomis macrochirus)<br>LC50: =11.5mg/L (96h, Lepomis macrochirus)<br>LC50: 34.09 - 47.64mg/L (96h, Poecilia reticulata)<br>LC50: =31mg/L (96h, Poecilia reticulata) | -                          | EC50: 4.24 - 10.7mg/L (48h, Daphnia magna)<br>EC50: 10.2 - 15.5mg/L (48h, Daphnia magna) |



|  |  |                                                                                                                                                                                       |  |  |
|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
|  |  | LC50: =27.8mg/L (96h, Brachydanio rerio)<br>LC50: =0.00175mg/L (96h, Cyprinus carpio)<br>LC50: 33.9 - 43.3mg/L (96h, Oryzias latipes)<br>LC50: 23.4 - 36.6mg/L (96h, Oryzias latipes) |  |  |
|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

#### Component Information

| Chemical name      | Partition coefficient |
|--------------------|-----------------------|
| Phenol<br>108-95-2 | 1.47                  |

**Other adverse effects** No information available.

### 13. Disposal considerations

#### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

### 14. Transport information

**TDG** Not regulated

**DOT** Not regulated

**MEX** Not regulated

**IATA** Not regulated

UN number or ID number UN2927

Packing group II

**IMDG** Not regulated

### 15. Regulatory information

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories**

Contact supplier for inventory compliance status

**16. Other information**

|                                   |                           |                                  |                           |                                           |
|-----------------------------------|---------------------------|----------------------------------|---------------------------|-------------------------------------------|
| <b>NFPA</b>                       | <b>Health hazards</b> 3   | <b>Flammability</b> 2            | <b>Instability</b> 0      | <b>Physical and chemical properties</b> - |
| <b>HMIS</b>                       | <b>Health hazards</b> 3 * | <b>Flammability</b> 2            | <b>Physical hazards</b> 0 | <b>Personal protection</b> X              |
| <i>Chronic Hazard Star Legend</i> |                           | <i>* = Chronic Health Hazard</i> |                           |                                           |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 Japan GHS Classification  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 RTECS (Registry of Toxic Effects of Chemical Substances)  
 World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 30-May-2024

**Revision Note** Reformatted and updated existing information.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**